A-A-55503
March 9, 1994

COMMERCIAL ITEM DESCRIPTION

MICROCIRCUIT, DIGITAL, 8-BIT NMOS MICROCONTROLLER WITH 2K X 8 EPROM AND 128K X 8 RAM

The General Services Administration has authorized the use of this commercial item description (CID).

1.0 Abstract. This CID covers the general requirements for all 8-bit nmos microcontroller with 2K x 8 EPROM and 128K x 8 RAM, digital microcircuit. Digital microcircuits covered by this CID are intended for commercial/industrial applications and shall not be used in military systems needing stringent environmental and electrical requirements.

1.1 Part or Identifying Number (PIN). The PIN for the CID shall be as shown in the following example:

A-A-55503	-	01
CID number		Dash number

2.0 Salient characteristics.

2.1 Maximum operating conditions. The maximum operating conditions shall be as specified in table I.

2.2 Electrical performance characteristics. The electrical performance characteristics shall be as specified in table II and figure 1.

2.3 Design and construction. Microcircuits supplied to this CID shall be as specified herein and on figure 2.

2.4 Programming characteristics. The programming characteristics shall be as specified in table III and figure 3.

2.5 Marking. Microcircuits supplied to this CID shall be marked with the manufacturer's standard commercial PIN.

3.0 Regulatory requirements. This section is not applicable to this CID.

AMSC N/A

FSC 5962

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TABLE I. Maximum operating conditions.

Ambient temperature under bias	0°C to 70°C
Storage temperature	-65°C to +150°C
Voltage on any pin with respect to ground	-.5 V to +7 V
Power dissipation	1.5 W

TABLE II. Electrical performance characteristics.

Test	Symbol	Conditions $0°C \leq T_C \leq +70°C$ unless otherwise specified $V_{CC} = V_{DD} = +5$ V ±10%	Device type	Limits		Unit
				Min	Max	
Input low voltage (except XTAL1, XTAL2, RESET)	V_{IL}		All	-0.5	0.8	V
Input low voltage (XTAL1, XTAL2, RESET)	V_{IL1}		All	-0.5	0.6	V
Input high voltage (except XTAL1, XTAL2, RESET)	V_{IH}		All	2.0	V_{CC}	V
Input high voltage (XTAL1, XTAL2, RESET)	V_{IH1}		All	3.5	V_{CC}	V
Output low voltage (D_0 - D_7)	V_{OL}	$I_{OL} = 2.0$ mA	All		0.45	V
Output low voltage (SYNC, P_{10} - P_{17}, P_{20} - P_{27})	V_{OL1}	$I_{OL} = 1.6$ mA	All		0.45	V
Output low voltage (PROG)	V_{OL2}	$I_{OL} = 1.0$ mA	All		0.45	V
Output high voltage (D_0 - D_7)	V_{OH}	$I_{OH} = -400$ µA	All	2.4		V
Output high voltage (all other outputs)	V_{OH1}	$I_{OH} = -50$ µA	All	2.4		V
Input leakage current (T_0, T_1, RD, WR, CS, A0, EA)	I_{IL}	$V_{SS} \leq V_{IN} \leq V_{CC}$	All		±10	µA
Output leakage current (D_0 - D_7, High Z state)	I_{OFL}	$V_{SS} + .45 \leq V_{OUT} \leq V_{CC}$	All		±10	µA
Low input load current (P_{10} - P_{17}, P_{20} - P_{27})	I_{LI}	$V_{IL} = 0.8$ V	All		0.3	mA
Low input load current (RESET, SS)	I_{LI1}	$V_{IL} = 0.8$ V	All		0.2	mA
V_{DD} supply current	I_{DD}	Typical = 5 mA	All		10	mA

See footnotes at end of table.

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TABLE II. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $0°C \leq T_C \leq +70°C$ unless otherwise specified $V_{CC} = V_{DD} = +5\ V \pm 10\%$	Device type	Limits		Unit
				Min	Max	
Total supply current	$I_{CC} + I_{DD}$	Typical = 60 mA	All		125	mA
Input leakage current (P_{10} - P_{17}, P_{20} - P_{27})	I_{IH}	$V_{IN} = V_{CC}$	All		100	µA
Input capacitance	C_{IN}		All		10	pF
I/O capacitance	$C_{I/O}$		All		20	pF
CS, A_0 setup to RD↓	t_{AR}		All	0		ns
CS, A_0 hold after RD↓	t_{RA}		All	0		ns
RD pulse width	t_{RR}		All	160		ns
CS, A_0 to data out delay	t_{AD}		All		130	ns
RD↓ to data out delay	t_{RD}		All		130	ns
RD↓ to data float delay	t_{DF}		All		85	ns
Cycle time 1/	t_{CY}		All	1.25	15	µs
CS, A_0 setup to WR↓	t_{AW}		All	0		ns
CS, A_0 hold after WR↓	t_{WA}		All	0		ns
WR pulse width	t_{WW}		All	160		ns
Data setup to WR↓	t_{DW}		All	130		ns
Data hold after WR↓	t_{WD}		All	0		ns
DACK to WR or RD	t_{ACC}		All	0		ns
RD or WR to DACK	t_{CAC}		All	0		ns
DACK to data valid	t_{ACD}		All		130	ns
RD or WR to DRQ cleared 2/	t_{CRQ}	$t_{CY} = 1.25\ \mu s$	All		100	ns
Port control setup before falling edge of PROG 3/	T_{CP}	$1/15\ t_{CY}-28$	All	55		ns

See footnotes at end of table.

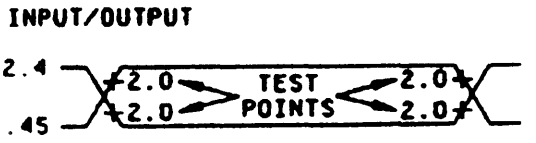
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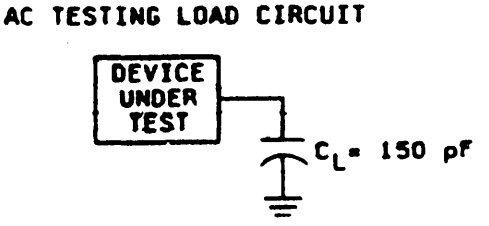
TABLE II. Electrical performance characteristics - Continued.

Test	Symbol	Conditions $0°C \leq T_C \leq +70°C$ unless otherwise specified $V_{CC} = V_{DD} = +5\ V \pm 10\%$	Device type	Limits		Unit
				Min	Max	
Port control hold after falling edge of PROG 4/	T_{PC}	$t_{CY} = 1.25\ \mu s$ $1/10\ t_{CY}$	All	125		ns
PROG to time P2 input must be valid 3/	T_{PR}	$t_{CY} = 1.25\ \mu s$ $8/15\ t_{CY}-16$	All		650	ns
Input data hold time 4/	T_{PF}		All	0	150	ns
Output data setup time 3/	T_{DP}	$t_{CY} = 1.25\ \mu s$ $2/10\ t_{CY}$	All	250		ns
Output data hold time 4/	T_{PD}	$t_{CY} = 1.25\ \mu s$ $1/10\ t_{CY}-80$	All	45		ns
PROG pulse width	T_{PP}	$t_{CY} = 1.25\ \mu s$ $6/10\ t_{CY}$	All	750		ns

1/ $T_{CY} = 15$ f/XTAL
2/ $C_L = 150$ pF
3/ $C_L = 80$ pF
4/ $C_L = 20$ pF

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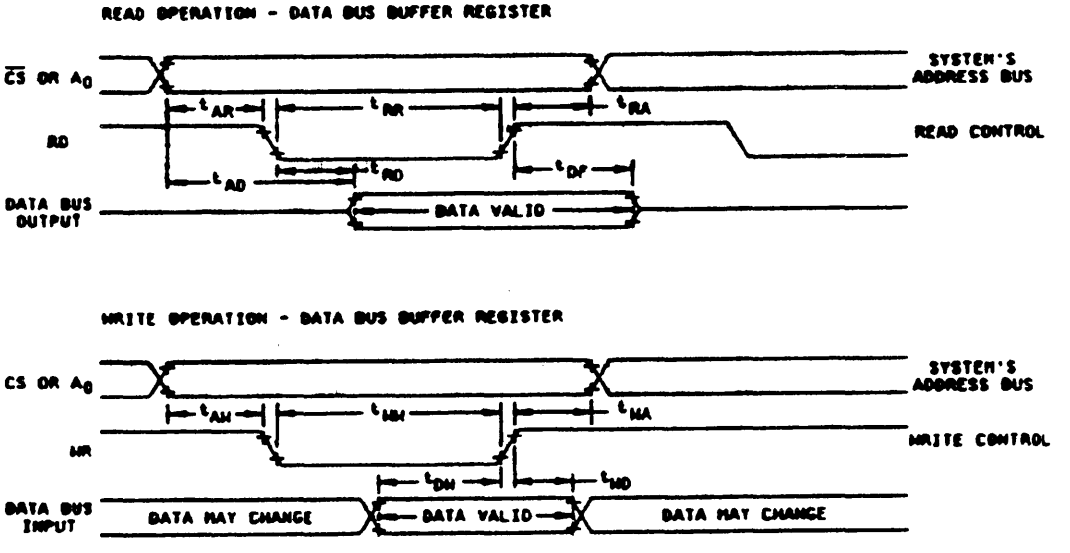
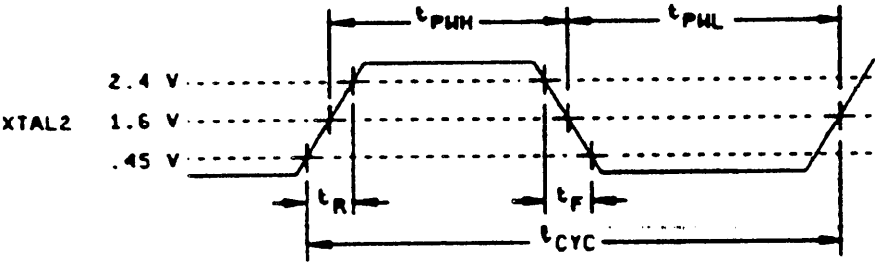


Figure 1. Output load circuit and timing waveforms.

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CLOCK TIMING



DMA

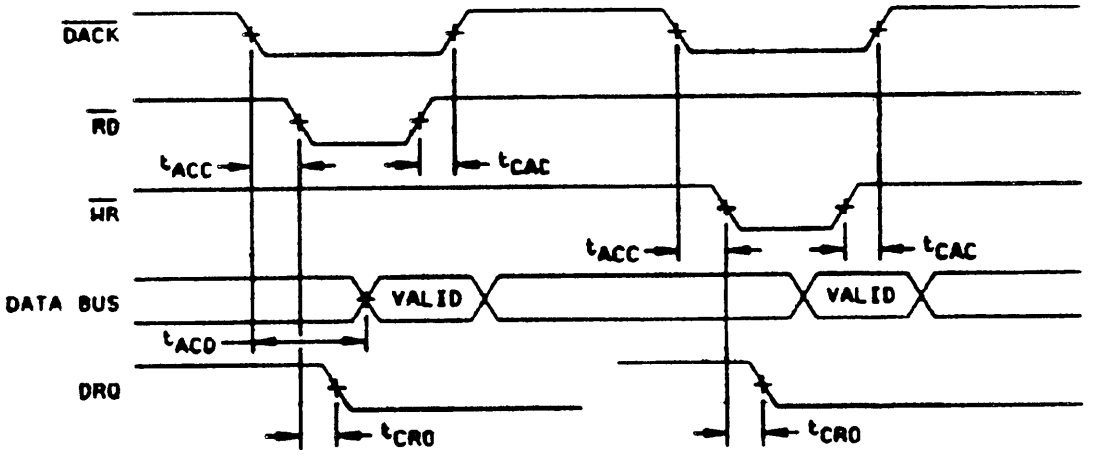
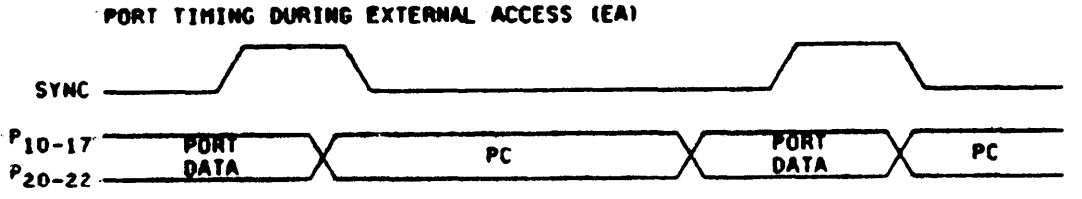


Figure 1. Output load circuit and timing waveforms - Continued.

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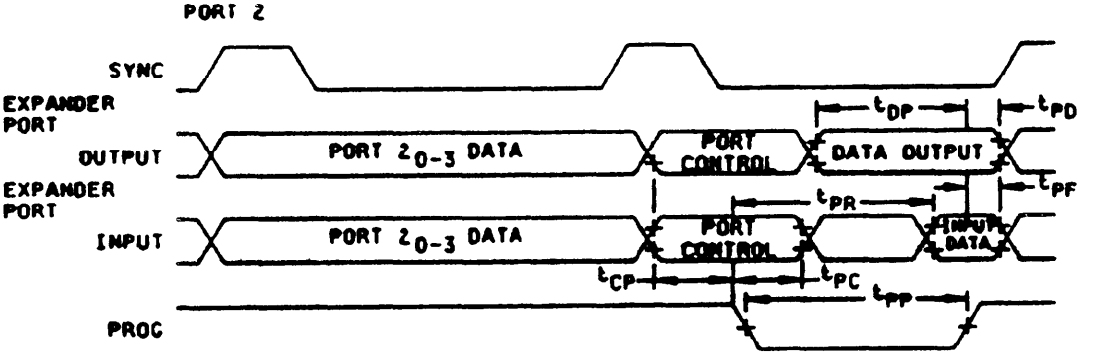
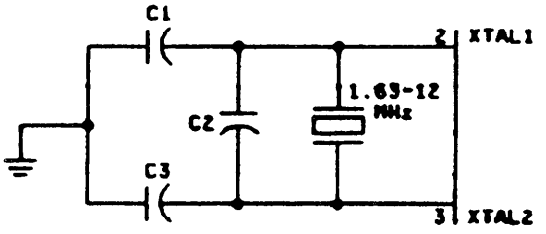


Figure 1. Output load circuit and timing waveforms - Continued.

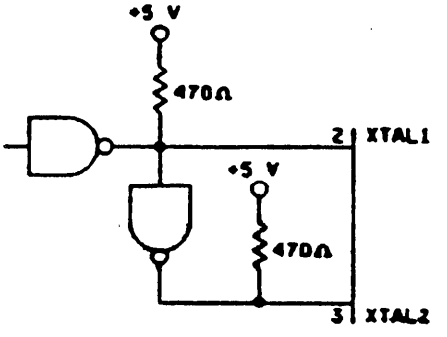
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CRYSTAL OSCILLATOR MODE



CRYSTAL SERIES RESISTANCE SHOULD BE
< 750Ω AT 12 MHz; < 180Ω AT 3.6 MHz.

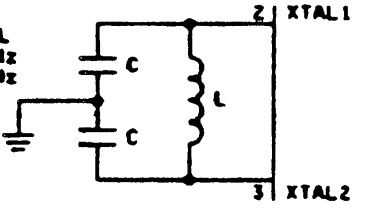
DRIVING FROM EXTERNAL SOURCE



RISE AND FALL TIMES SHOULD NOT EXCEED 20 ns.
RESISTORS TO V_{CC} ARE NEEDED TO ENSURE
V_{IH} = 3.5 V IF TTL CIRCUITRY IS USED.

LC OSCILLATOR MODE

L	C	NORMAL
45 H	20 pF	5.2 MHz
120 H	20 pF	5.2 MHz



$$f = \frac{1}{2\pi\sqrt{LC}}$$

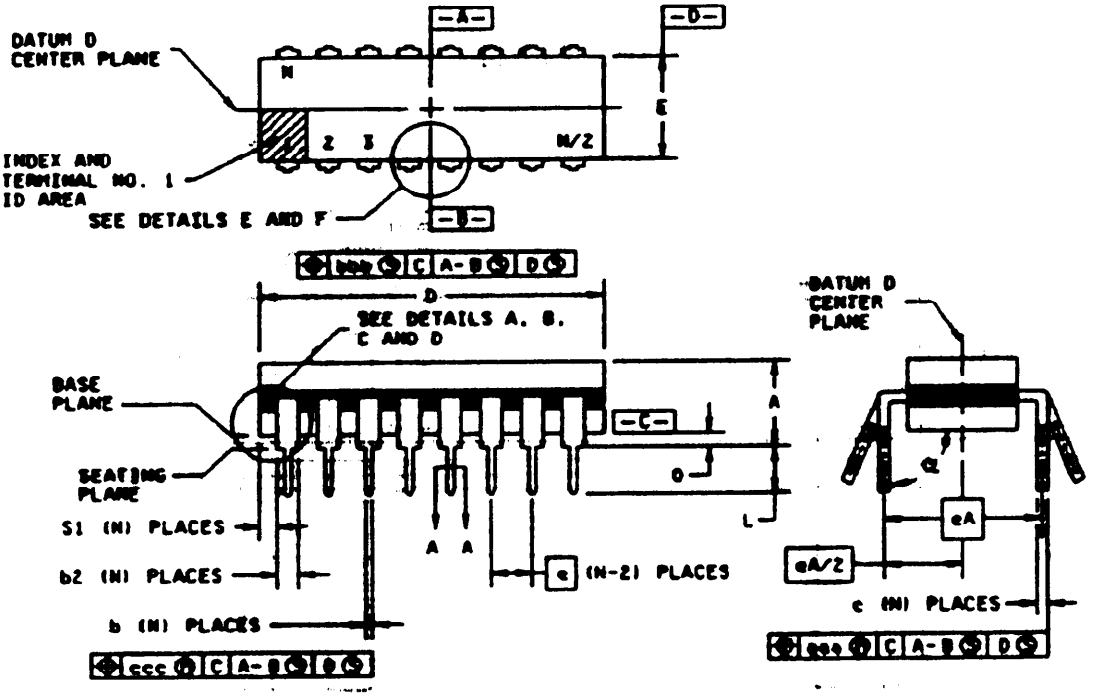
$$C = \frac{C + 3\,C_{pp}}{2}$$

C_{pp} = 5 pF-10 pF
PIN-TO-PIN
CAPACITANCE

EACH C SHOULD BE APPROXIMATELY 20 pF, INCLUDING STRAY CAPACITANCE

Figure 1. Output load circuit and timing waveforms - Continued.

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Symbol	Min	Max
A	---	5.72 (.225)
b	0.36 (.014)	0.66 (.026)
b1	0.36 (.014)	0.58 (.023)
b2	1.14 (.045)	1.65 (.065)
b3	0.58 (.023)	1.14 (.045)
c	0.20 (.008)	0.46 (.018)
c1	0.20 (.008)	0.38 (.015)
D	---	53.24 (2.096)
E	12.95 (.510)	15.75 (.620)
E2	7.11 (.280)	---
E3	1.27 (.050)	---
e	2.54 (.100) BSC	

Symbol	Min	Max
eA	7.62 (.600) BSC	
eA/2	3.81 (.300) BSC	
L	3.18 (.125)	5.08 (.200)
Q	0.38 (.015)	1.78 (.070)
α1	0.51 (.020)	---
S1	0.13 (.005)	---
S2	0.13 (.005)	---
α	90°	105°
aaa	---	0.38 (.015)
bbb	---	0.76 (.030)
ccc	---	0.25 (.010)
M	---	0.038 (.0015)
N	40	

NOTES:
1. Dimensions are in millimeters.
2. Inches are in parentheses.
3. The US government preferred system of measurement is the metric SI system. However, this item was originally designed using inch-pound units of measurement. In the event of conflict between the metric and inch-pound units, the inch-pound units shall take precedence.

Figure 2. Design and construction.

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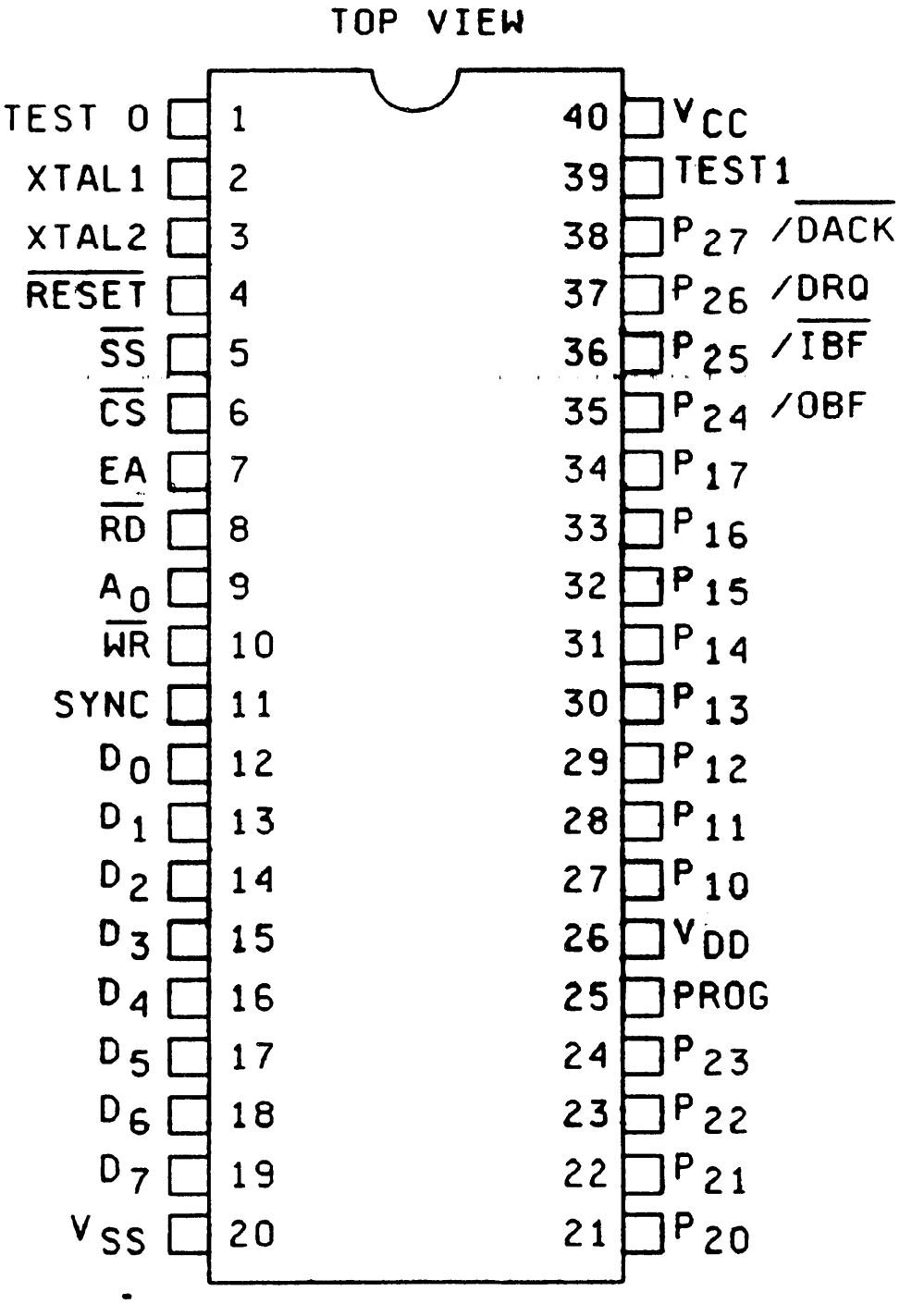


Figure 2. Design and construction - Continued.

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TABLE III. Programming characteristics.

Symbol	Parameter	Limits		Units
		Min	Max	
	DC characteristics			
V_{DDH}	V_{DD} program voltage high level	20.5	21.5	V
V_{DDL}	V_{DD} voltage low level	4.75	5.25	V
V_{PH}	PROG program voltage high level	17.5	18.5	V
V_{PL}	PROG voltage low level	V_{CC} - .5	V_{CC}	V
V_{EAH}	EA program or verify voltage high level	17.5	18.5	V
V_{EAL}	EA voltage low level		5.25	V
I_{DD}	V_{DD} high voltage supply current		30	mA
I_{PROG}	PROG high voltage supply current		1.0	mA
I_{EA}	EA high voltage supply current		1.0	mA
	AC characteristics			
t_{AW}	Address setup time to reset ↑	$4t_{CY}$		
t_{WA}	Address hold time after reset ↑	$4t_{CY}$		
t_{DW}	Data in setup time to PROG↓	$4t_{CY}$		
t_{WD}	Data in hold time after PROG↓	$4t_{CY}$		
t_{PH}	RESET hold time to verify	$4t_{CY}$		
t_{VDDW}	V_{DD} setup time to PROG↓	0	1.0	ms
t_{WDDH}	V_{DD} hold time after PROG↓	0	1.0	ms
t_{PW}	Program pulse width	50	60	ms
t_{TW}	Test 0 setup time for program mode	$4t_{CY}$		
t_{WT}	Test 0 hold time after program mode	$4t_{CY}$		
t_{DO}	Test 0 to data out delay		$4t_{CY}$	
t_{WW}	RESET pulse width to latch address	$4t_{CY}$		
t_r, t_f	V_{DD} and PROG rise and fall times	.5	2.0	μs
t_{CY}	CPU operation cycle time	4.0		μs
t_{RE}	RESET setup time before EA↓	$4t_{CY}$		

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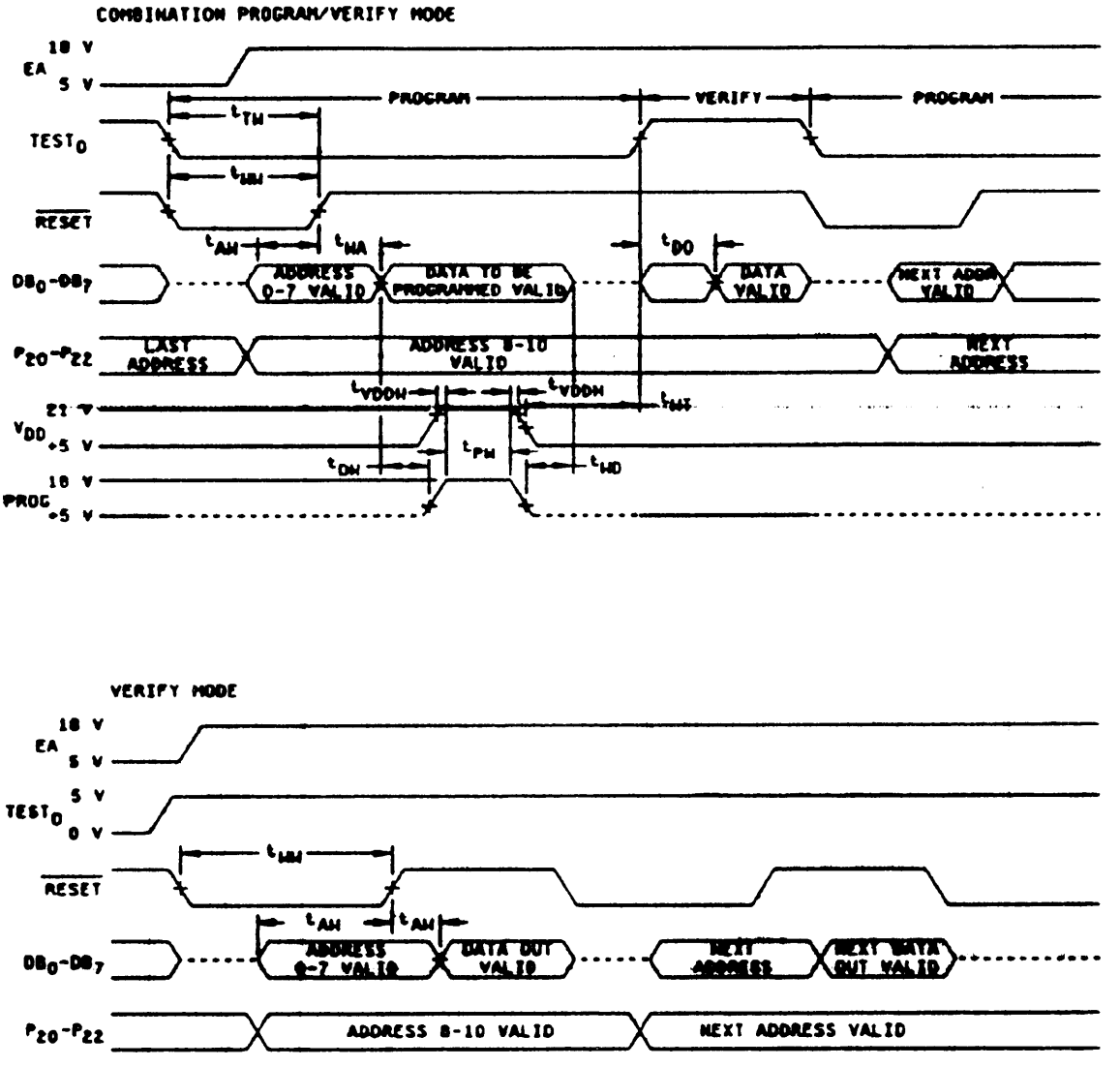


Figure 3. Programming waveforms.

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4.0 Quality assurance provisions.

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection, examination, and test requirements specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections, examinations, or tests set forth in this description where such inspections, examinations, and tests are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Contractor certification statement. The contractor shall certify and maintain objective quality evidence that the product offered meets the requirements of this CID, and that the product conforms to the producer's own drawings, specifications, standards, quality assurances practices, and is the same as the product provided as a bid sample. The acquiring activity reserves the right to require proof of such conformance prior to the first delivery and thereafter as may be otherwise provided for under the provisions of the contract.

4.3 Certificate of conformance. A certificate of conformance shall accompany all microcircuits supplied to this CID.

5.0 Packaging.

5.1 Preservation, packaging, packing, labeling, and marking. Preservation, packaging, labeling, and marking shall be as specified in the contract or purchase order.

6.0 Notes. This section contains relevant information which is useful to buyers, users, and suppliers in the process of acquiring the item, but is not mandatory.

Pin descriptions.

Symbol	Pin	Type	Name and function
TEST 0, TEST 1	1, 39	I	Test inputs: Input pins which can be directly tested using conditional branch instructions. Frequency reference: TEST 1 ($\overline{T_1}$) also functions as the event timer input (under software control). TEST 0 (T_0) is used during PROM verifications and EPROM verification.
XTAL 1, XTAL 2	2, 3	I	Inputs: Inputs for a crystal, LC or an external timing signal to determine the internal oscillator frequency.
$\overline{\text{RESET}}$	4	I	RESET: Input used to reset status flip-flops and to set the program counter to zero. RESET is also used during EPROM programming and verification.
$\overline{\text{SS}}$	5	I	Single step: Single step input used in conjunction with the SYNC output to step the program through each instruction (EPROM). This should be tied to +5 V when not used.
$\overline{\text{CS}}$	6	I	Chip select: Chip select input used to select one UPI microcomputer out of several connected to a common data bus.
EA	7	I	External access: External access input which allows emulation, testing and EPROM verification. This pin should be tied low if unused.
$\overline{\text{RD}}$	8	I	Read: I/O read input which enables the master CPU to read data and status words from the output data bus buffer or status register.
$\overline{\text{WR}}$	10	I	Write: I/O write input which enables the master CPU to write data nd command words to the UPI input data bus buffer.
A_0	9	I	Command/Data select: Address input used by the master processor to indicate whether byte transfer is data (A_0 = 0, F1 is RESET) or command (A_0 = 1, F1 is set). A_0 = 0 during program and verify operations.

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Symbol	Pin	Type	Name and function
SYNC	11	O	Output clock: Output signal which occurs once per UPI instruction cycle. SYNC can be used as a strobe for external circuitry; it is also used to synchronize single step operation.
D_0 - D_7 (bus)	12-19	I/O	Data bus: Three-state, bidirectional data bus buffer lines used to interface the UPI microcomputer to an 8-bit master system data bus.
P_{10} - P_{17}	27-34	I/O	Port 1: 8-bit, Port 1 quasi-bidirectional I/O lines.
P_{20} - P_{27}	21-34 35-38	I/O	Port 2: 8-bit, Port 2 quasi-bidirectional I/O lines. The lower 4 bits (P_{20} - P_{23}) interface directly to the 8243 I/O expander device and contain address and data information during port 4-7 access. The upper 4 bits (P_{24} - P_{27}) can be programmed to provide interrupt request and DMA handshake capability. Software control can configure P_{24} as output buffer full (OBF) interrupt, P_{25} as input buffer full (IBF) interrupt, P_{26} as DMA request (DRQ), and P_{27} as DMA acknowledge (DACK).
PROG	25	I/O	Program: Multifunctional pin used as the program pulse input during PROM programming. During I/O expander access the PROG pin acts as an address/data strobe to the 8243. This pin should be tied high if unused.
V_{CC}	40		Power: +5 V main power supply pin.
V_{DD}	26		Power: +5 V during normal operation. +21 V during programming operation. Low power standby supply pin.
V_{SS}	20		Ground: Circuit ground potential.

6.1 Ordering data. The contract or purchase order shall specify the following:

a. CID document number and revision and CID PIN.
b. Quality assurance provisions.
c. Packaging requirements.

6.2 Comments. Comments on this CID should be directed to Defense Electronics Supply Center, 1507 Wilmington Pike, ATTN: DESC-ECC, Dayton, OH 45444-5270, or telephone (513) 296-8526.

6.3 Sources of supply. Sources of supply are listed herein. Additional sources will be added as they become available.

CID dash number	Vendor Cage number	Vendor Commercial PIN
01	34649	D8742
02	34649	D8742 1/

1/ When the 02 device is specified, the 01 device may be used interchangeably.

Vendor CAGE number
34649

Vendor name and address
Intel Corporation
3065 Bowers Avenue
Santa Clara, CA 95051

CIVIL AGENCY COORDINATING ACTIVITY:

GSA - 7FXE

PREPARING ACTIVITY:
DLA-ES

(Project 5962-D728)

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